CellVerity — User Guide

Formerly ICC CTCF Ölçer

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Source code and downloads: https://github.com/erdal-b-dot/cellverity · Türkçe kılavuz: `KULLANIM_KILAVUZU.md`

Contents

1. What does the program do?
2. Installation
3. Image requirements
4. Quick start (5 minutes)
5. The main window
6. Measuring a single image — step by step
7. Editing cell boundaries
8. Batch-measuring a folder
9. Results, Excel and autosave
10. Plots and statistics (comparing groups)
11. Fiji bridge and method validation
12. Warnings: what they mean and what to do
13. For your paper: methods text and citation
14. Troubleshooting
15. Technical details

1. What does the program do?

It measures **per-cell CTCF** (corrected total cell fluorescence) in immunocytochemistry (ICC) fluorescence images:

> **CTCF = Integrated density – (Cell area × Mean background)** (McCloy et al., 2014, *Cell Cycle* 13:1400–1412)

It automates the manual ImageJ workflow:

Manually in ImageJ	**In this program**
Calibrate with Set Scale	The scale bar **and the value printed under it** (20 µm, 50 µm) are read automatically
Trace every cell with the polygon tool	Cells are segmented from FITC; touching cells are separated using DAPI nuclei
Draw and measure 3 background regions	3 boxes are placed automatically away from cells (movable, or add your own)
Measure → copy to Excel → formula	Table, Excel, group summary, plots and statistics in one click

Measurements are identical to ImageJ/Fiji. For the same cell boundaries, Area, Mean, StdDev, Min, Max, IntDen and RawIntDen are the same as Fiji's (largest difference < 0.00001 %). You can check this yourself from within the program: **Fiji ▸ Verify with Fiji** (Section 11).

Your images are **never sent anywhere**; the program works offline.

Language: the interface follows your computer's language (Turkish or English). Change it under **View ▸ Language / Dil** (takes effect after restart). Excel output is always in English.

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2. Installation

2.1 Ready-to-run app (recommended — no Python needed)

Download from **https://github.com/erdal-b-dot/cellverity/releases** (newest version at the top).

Computer	File	Minimum system
Apple Silicon Mac (M1/M2/M3/M4)	`CellVerity_<version>_mac_AppleSilicon.dmg`	macOS 12.3
Intel Mac	`CellVerity_<version>_mac_Intel.dmg`	macOS 12.0 (Monterey)
Windows 10/11 (64-bit)	`CellVerity_<version>_Windows.zip`	Windows 10
Linux x86_64 (from v2.2.0)	`CellVerity_<version>_Linux_x86_64.AppImage` (or `.tar.gz`)	Ubuntu 22.04, Debian 12, Fedora 36 (glibc 2.35)
Linux ARM64 (from v2.2.0)	`CellVerity_<version>_Linux_aarch64.AppImage` (or `.tar.gz`)	glibc 2.39 (e.g. Ubuntu 24.04)

> To find your Mac's processor: Apple menu ▸ **About This Mac**. "Chip: Apple M..." means Apple Silicon; "Processor: Intel..." means Intel.

macOS: 1. Double-click the `.dmg` → drag **CellVerity** onto the **Applications** folder shown next to it. Then eject the disk (⏏) in the Finder sidebar. 2. **On first launch** do not double-click: **right-click the app ▸ Open ▸ Open**. (The app is not notarized by Apple, so macOS shows an "unidentified developer" warning; this is needed only once.) 3. If you see *"is damaged and can't be opened"* (happens with downloaded files), run once in Terminal: `xattr -cr "/Applications/CellVerity.app"`

Windows: 1. Right-click the zip ▸ **Extract All**. Put the folder e.g. in `Documents` (do not run it from inside the zip). 2. Double-click `CellVerity.exe`. If *"Windows protected your PC"* appears: **More info ▸ Run anyway**. 3. Desktop shortcut: right-click the exe ▸ Send to ▸ Desktop (create shortcut).

Linux (from v2.2.0): 1. Put the `.AppImage` in a permanent folder, e.g. `~/Applications/CellVerity.AppImage`. 2. In a terminal, make it executable once and run it: `chmod +x ~/Applications/CellVerity.AppImage` → `~/Applications/CellVerity.AppImage` (Type the first letters and press **Tab** instead of typing the name; a typo gives "No such file or directory".) 3. If you see a "FUSE" / `libfuse.so.2` error: `sudo apt install libfuse2` (Ubuntu 22.04) or `libfuse2t64` (24.04). Otherwise extract the `.tar.gz` and run `./CellVerity/CellVerity` (no FUSE needed). 4. To add it to

the application menu, create `~/.local/share/applications/cellverity.desktop`: `[Desktop Entry] Type=Application Name=CellVerity Exec=/home/<user>/Applications/CellVerity.AppImage Icon=applications-science Terminal=false Categories=Science;` Then search *CellVerity* in **Activities**; right-click ▸ **Add to Favorites**. For a new version, replace the AppImage under the same name.

Updates: at startup (at most once a day) the program only asks GitHub *which is the latest version?*; if a new version exists, a **Download / Skip this version / Later** dialog appears. No images or data are sent. Turn it off with **Help ▸ Check for updates at startup**; check manually with **Help ▸ Check for updates...**. Your version and build: **Help ▸ About / version info** (use *Copy info* when reporting a problem).

Appearance: View ▸ Theme — *Dark* (default) or *System* (the operating system's light/dark appearance); applied at the next start. The area around the image stays dark in both. Only three icons are coloured: **Open FITC** green, **Open DAPI** blue, **Delete** red. When the window is narrow, toolbars hide their labels and show icons only (the label appears on hover).

2.2 Running from source

With a Python environment (to receive updates via `git pull`): - macOS: `./install_mac.sh`, then `./run_desktop.command` - Windows: `install_windows.bat`, then `start_desktop.bat`

Building the stand-alone packages yourself: `./build_mac.sh` (macOS) / `build_windows.bat` (Windows).

2.3 Fiji (optional but recommended)

The Fiji bridge (Section 11) requires Fiji: https://imagej.net/software/fiji/downloads The program looks for Fiji in `Downloads/Fiji`, `Applications/Fiji`, your home folder (`~/Fiji`, `C:\Fiji`). Elsewhere: **Fiji ▸ Choose Fiji location...** (asked once, remembered).

2.4 Earlier browser version (v1)

v1 (the browser-based Streamlit version, Turkish only) is still available under the `v1.0.0` tag. v1 and v2 give the same numbers for the same cell boundaries, but the **nucleus detection method differs**, so boundaries between touching cells are drawn slightly differently (Section 15). Measure all images of a study with **the same version**.

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3. Image requirements

However good the program is, the validity of the measurement depends on the image.

Required: - **Raw, single-channel images.** The FITC image saved directly by the microscope software (cellSens etc.). Images that went through Photoshop, had brightness/contrast adjusted, or were **merged cannot be used** for CTCF — the program detects these and warns you. - **All compared images acquired with identical settings:** same objective, **exposure time**, gain, lamp power, white balance. Fluorescence signal scales with exposure: a cell in a 1 s image looks ~2× brighter than in a 0.5 s image. Turn off auto-exposure in cellSens. The program reads the exposure time from the image (EXIF), shows it in the panel and in Excel, and warns when exposures are mixed. - **Same objective.** Do not compare 20× with 40× images — the program warns if objectives are mixed. - **No saturated pixels.** The number of saturated pixels is reported for every cell ("Saturated px" column).

Recommended: - **The DAPI image of the same field.** Essential for separating touching cells correctly (Warnings). - **Calibration:** the image needs a scale bar (cellSens style: white box in the lower right corner) **or** pixel-size information in the file (cellSens raw TIFF, OME-TIFF, ImageJ TIFF). Without either, you enter the pixel size manually (6.2). - **TIFF** is better than JPEG (JPEG is lossy). Use the same format for all **FITC images** of a study (DAPI is only used to separate cells and may be JPG; TIFF FITC + JPG DAPI pairing is supported). - **Supported files:** JPG, PNG, TIFF (8/16-bit, RGB or single channel). Vendor formats (.czi, .nd2, .lif, .vsi) are not supported yet; save them as TIFF in Fiji (the pixel size is kept).

Tested equipment: Olympus BX43 fluorescence microscope, Olympus DP74 colour camera, cellSens Entry software. On this camera FITC appears yellow-green and DAPI **cyan**. FITC is measured on the **green (G) channel**; FITC and DAPI are told apart by the blue/red ratio of the brightest pixels. Other cameras/software should work for measurement; if the scale-bar label cannot be read, the program warns you and asks for the value (Section 6.2).

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4. Quick start (5 minutes)

> Example images to try: the `examples/` folder in the repository (two FITC + DAPI pairs with expected results, CC BY 4.0).

1. Open the program.
2. In Finder/Explorer, **select the FITC and DAPI files together** and drag them onto the window. The program tells which one is DAPI by its colour.
3. Check the right panel: **Calibration** (e.g. 0.1449 µm/pixel, 20 µm, *read from label*), **Objective**, **Cell count**, the green **"Cell separation: DAPI nuclei"** box.
4. **Select all cells** (or click the cells to measure — they turn yellow). The **Cells** tab below shows CTCF.
5. In the right panel enter **Group / condition** (e.g. *Control*) and **Experiment / replicate** (e.g. *1*) → ➕ **Add to results table** (⌘↩ / Ctrl+Enter).
6. Next image: close with **⌘W**, drag the next pair, change the group, add.
7. When done: **Save to Excel** (⌘E) and **Plot…** (⌘G).

With many images, use **Batch-measure folder** (⌘B) instead of steps 4–6 (Section 8).

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5. The main window

Part	Contents
Menus	File · Results · Fiji · Edit · View · Help
Toolbar 1	Open FITC · Open DAPI · Close images · Fit/Zoom in/Zoom out · **Display:** FITC / DAPI / Merged · Nucleus outlines
Toolbar 2	**Boundary editing:** Select · Split · Merge · Delete · Draw · Undo · Redo — **Results:** add to table · batch · Excel · Plot · Verify with Fiji
Image area	Mouse wheel: zoom · Drag: pan · Click: select cell
Right panel	Files, scale bar, calibration, objective, counts, cell-separation indicator, cell selection, background settings, results (group, experiment), legend, display brightness
Bottom tabs	Cells · Background boxes · Collected results · Group summary

Outlines on the image:

Outline	Meaning
Grey, numbered	Cell boundary found by the program
Yellow	Selected (measured) cell
Magenta	Nucleus boundary found in DAPI
Red square (BG1, BG2…)	Background box

The **Display brightness** slider and **Display: FITC/DAPI/Merged** change **only the screen**; measurement always uses the raw FITC (green channel) data.

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6. Measuring a single image — step by step

6.1 Opening images

- **Drag and drop** (easiest): drag FITC + DAPI together.
- **Open FITC... (⌘O)** and **Open DAPI... (⌘D)**.
- If you use the wrong button (e.g. a blue file with "Open FITC"), the program asks: *"... looks like a DAPI image, open it as DAPI?"*
- **Opening a new FITC removes the previous field's DAPI** (another field's nuclei would separate cells wrongly). Open the new field's DAPI too.
- **Close images (⌘W):** clears the window; collected results are kept.

6.2 Checking the calibration

The **Calibration** line in the right panel also tells the source: - *read from label* — the value under the scale bar (20 µm, 50 µm) was read. ✅ - *entered manually* — the value you typed into the **Scale bar** box. - *label NOT READ, assumed* — a red warning appears; type the value shown in the image into the **Scale bar** box. - *from file metadata (OME / Olympus SIS / ImageJ)* — the image has no scale bar; the pixel size was read from the file. cellSens **raw TIFF** files are like this (no bar burned in, pixel size stored in the file). ✅ - *entered manually* (µm/pixel) — **Edit ▸ Enter calibration manually**: enter the pixel size directly or from the objective + camera pixel size (camera pixel ÷ magnification; DP74: 5.86 µm).

Order of precedence: manually entered value → scale bar in the image → file metadata. If both a bar and metadata exist and differ by more than 3 %, a red warning appears (the image may have been resized or the bar misread).

If no scale is found at all (no bar, no metadata) the program asks for the calibration in a dialog. With *Measure in pixels*, areas are in pixels² and the cell/nucleus size filters work in pixels — cells may be fragmented; use it only to compare images acquired with identical settings among themselves. In batch mode, uncalibrated images are flagged in the results and statistics.

> ⚠️ A wrong scale value makes **area and IntDen wrong several-fold** (e.g. 20 µm instead of 50 µm: 6.25×). Glance at the calibration line for every new image.

Known values for DP74 + cellSens: **20 µm = 138 px → 0.1449 µm/px (40×)**; **50 µm = 172 px → 0.2907 µm/px (20×)**.

6.3 Cell-separation indicator

Indicator	Meaning	What to do
🟢 **DAPI nuclei**	Touching cells were separated using nuclei	—
🟡 **Shape only (no touching cells)**	No DAPI, but none needed	—
🔴 **Shape only — N touching cell pairs**	Touching cells separated by shape only (~5% error)	Open the DAPI of the same field
🔴 **DAPI does not seem to belong to the same field**	Most nuclei lie outside the cells	Open the correct DAPI (consecutive numbers can still be different fields)

6.4 Selecting cells

- **Click** a cell → it turns yellow and is measured; click again → deselected.
- **Select all cells** / **Clear selection**.
- Do not measure cells **cut off** at the image border (their area is incomplete).

6.4.1 Quality control and documented exclusion

- **Automatic quality control (QC)** — *Exclude cells failing quality control* in the right panel (on by default). These cells are outlined in **orange** and are not selected by *Select all cells*:
- cells with **saturated pixels** (CTCF underestimated),
- cells with **no nucleus** or **several nuclei** (usually a segmentation error → Split/Merge),
- **partial nucleus**: nuclear area inside the cell < 30% of the image median (a cut-off or wrongly separated cell fragment),
- **pyknotic nucleus**: nuclear area < 50% of the median **and** DAPI intensity > 1.5× the median (a **heuristic** criterion for apoptotic nuclei; check visually).
- **Manual exclusion** — **right-click** a cell ▸ choose a reason: *apoptotic / pyknotic, mitotic, overlapping / not separable, out of focus, debris / not a cell, other*. The cell is outlined in **red**. To undo: right-click ▸ *Remove exclusion* (or click to select the cell).
- To measure a QC-flagged cell anyway, click to select it; its flag stays in the `QC_flags` column.
- Every excluded cell is written with its reason to the **Excluded_cells** sheet in Excel, with counts in **Exclusion_summary** — report as "n cells were excluded (x saturated, y apoptotic …)".
- Recommended approach to cell selection and sampling: predefined criteria, systematic field selection, exclusions made blind to treatment group, ≥ 3 independent experiments.

6.5 Background

- **Automatic** (default): **3 boxes** (50×50 pixels) ≥ 40 pixels away from cells, free of signal/debris, as close to the cells as possible. Size and number are set in the right panel.
- **Moving boxes:** hover over a box (hand cursor), press and drag. It is re-measured when released. Moving an automatic box switches to **Manual**; the other boxes stay in place.
- **Manual:** click an empty area → a box is added; click a box → it is removed. **Clear manual boxes**.

- A box overlapping a cell is not accepted (it moves back).
- Per-box values: **Background boxes** tab below.

> ℹ️ **If the background is 0**, the camera/cellSens clipped the dark background to zero; CTCF = IntDen. Comparisons are valid if all images were acquired with the same settings.

6.5.1 Nucleus / cytoplasm (N/C ratio)

If a DAPI image is open, the program measures the **nucleus** and the **cytoplasm** of every cell separately. Nothing else is needed; the results appear in the **N/C** column. Translocation of a protein into the nucleus (e.g. nuclear accumulation of p53 on activation) is a direct read-out of pathway activity. Whole-cell CTCF does not show it; the N/C ratio does.

- **Nucleus** = DAPI mask. **Cytoplasm** = cell − nucleus.
- **Edge ring** (right panel ▸ *Nucleus / cytoplasm (N/C)*, default **0.5 µm**): FITC and DAPI are acquired with separate exposures, so the nuclear edge can be shifted by a few pixels. The nucleus is shrunk by this distance and the cytoplasm starts this far from the nucleus. The ring belongs to neither compartment. 0 = no ring.
- **N/C ratio** = (nuclear mean − background) / (cytoplasmic mean − background). It is the ratio of background-corrected mean intensities and does not depend on cell size. Above 1, the signal is more concentrated in the nucleus.
- **Nuclear fraction of signal** (`Nuc_fraction`) = nuclear CTCF / (nuclear + cytoplasmic CTCF).
- **Thin cytoplasm:** if the measured cytoplasm is smaller than **20 %** of the cell area, the ratio comes from few pixels and is unreliable. The cell is flagged (*N/C QC* column: `thin_cytoplasm`) and left out of the N/C analysis **only**. Its whole-cell CTCF remains valid. The statistics text states how many cells were left out.
- In plots and statistics, **NC_ratio**, **Nuc_fraction**, **Nuc_CTCF** and **Cyto_CTCF** can be chosen as the *Variable*.

> ⚠️ **On a widefield microscope**, out-of-focus cytoplasm above and below the nucleus adds to the nuclear value, so the N/C ratio is biased toward 1. **Compare conditions (control vs treatment), not absolute values.** Confocal images are much less affected.
>
> ℹ️ Because it is a ratio of two measurements, the N/C ratio is largely independent of exposure time (provided the camera is linear). Acquiring the compared images with identical settings is still the safest approach.

6.6 Adding to the results table

- **Group / condition:** e.g. *Control*, *Drug 10 µM 24 h*. Plots and statistics group by this label.
- **Experiment / replicate:** independent experiment (biological replicate): *1*, *2*, *3*... Needed for statistics on experiment means (Section 10).

- ➕ **Add to results table (⌘↩).** Re-adding the same image replaces the earlier entry (it is not counted twice).

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7. Editing cell boundaries

Automatic boundaries are good in most images; dense clusters, debris or dim cells may need corrections. **Toolbar 2 ▸ Boundary editing**:

Tool	How	When
Select	Normal mode	Select cells, move boxes, pan
Split	Draw a line across touching cells **from one edge to the other**	Two (or more) cells were counted as one
Merge	Click one cell, then its neighbour	One cell was split in two
Delete	Click the region	Debris or a non-cell region
Draw	Draw the cell boundary with the mouse; it closes when released	A missed cell or a poorly outlined boundary (the drawing overrides cells beneath)
Undo / Redo	⌘Z / ⇧⌘Z (Windows: Ctrl+Z / Ctrl+Y), 30 steps	Every edit can be undone

- In Split and Draw modes, pan by holding **⌥ Option (Windows: Alt)** while dragging.
- Splitting and merging **preserve the total area**.
- Edits are written to the **Manual_edits** column in Excel (e.g. *merge 1+2; split 1*).
- Changing the scale value re-analyses the image and **resets the edits** (a warning appears below). Calibrate first, then edit.

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8. Batch-measuring a folder

Results ▸ Batch-measure folder... (⌘B)

8.1 Recommended folder layout

```
Experiments/
  Exp_1/
    Control/      ← FITC + DAPI images
    Treatment1/
    Treatment2/
  Exp_2/
    Control/
    ...
```

- **Group** = name of the folder containing the image; **Experiment** = the folder above it.
- If you select a single experiment folder (e.g. `Exp_1`), its name becomes the experiment name.
- Both can be edited in the table before measuring.

8.2 FITC–DAPI pairing

Each FITC is paired with the DAPI, among those **within ±3 file numbers in the same folder**, whose **nuclei actually overlap the cells**. Overlap is a **chance-corrected** score (1 = all nuclei inside cells, 0 = no better than random placement); ≥ 0.5 is required. A raw fraction would mislead: in a crowded field where cells cover half the image, a DAPI from another field still overlaps ~50%. Consecutive numbers do not always mean the same field — the program checks the images themselves.

Status	Meaning
🟢 **ready**	Pair found (percentage in the Overlap column; the **Exposure** column shows the FITC exposure time — a red warning appears below if exposures differ within the folder)
🟠 **no DAPI**	No matching DAPI; cells will be separated by shape only
⚪ **skipped: no camera information**	Merged or re-saved image
⚪ **skipped: processed image**	Saved by Photoshop etc.
⚪ **skipped: could not tell FITC from DAPI**	Ambiguous colour (e.g. merged)

Status	Meaning
⚪ **skipped: unpaired DAPI**	A DAPI that belongs to no FITC

8.3 Measurement

- Ticked rows are measured; untick those you do not want.
- In each image: **all cells** + **automatic background** (box size/number from the right panel).
- With *Exclude cells failing quality control* ticked, QC-flagged cells are not measured; together with border cells they are written with their reason to **Excluded_cells** (`border` , `saturated` , `no_nucleus` , `partial_nucleus` , `multinucleate` , `pyknotic`).
- **Cells touching the image border** are excluded by default.
- The **Selection** column in Excel records *manual click* or *batch: all cells, edge excluded*.
- After batch measurement, check suspicious images (especially dense clusters): in the **Collected results** tab, **hover over a row** to preview the cell; **click the row** to open that image with its DAPI — the cells recorded in the table are selected and the clicked cell is centred. Edit boundaries (Section 7) or **Verify with Fiji**, then **Add to results table** to replace that image's earlier rows.

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9. Results, Excel and autosave

9.1 Excel sheets

Sheet	Contents
Cells	All per-cell measurements with context
Group_summary	n, mean, SD, SEM, median, min, max per group (CTCF)
Image_means	Image (field) means — ready to paste into GraphPad
Experiment_means	Experiment means — the recommended statistical unit
Background	Per-box background measurements
Excluded_cells, **Exclusion_summary**	Excluded cells and their reasons (automatic QC / manual)
Methods	Formulas, methods, citation
Validation, **Validation_summary**	Manual (Fiji) vs automatic comparison and Bland–Altman summary (if present)

9.2 Key columns in the Cells sheet

Column	Meaning
Protein	Protein / target measured (10.2)
Area, Mean, StdDev, Min, Max, IntDen, RawIntDen	Same definitions as ImageJ
BG_Mean, CTCF, CTCF_per_Area	Background mean, CTCF, CTCF/area (= Mean − BG_Mean)
Saturated_px	Saturated pixel count (should be 0)
um_per_px, Scale_bar_um, Scale_source, Objective	Calibration and its source (label / user / default)
Exposure_s, ISO	Exposure time and gain (from the image EXIF; must be identical across compared images)
BG_mode, BG_box_px, BG_n	Background method
Segmentation, Nucleus_method, DAPI_overlap, Touching_pairs	Cell separation details

Column	Meaning
Selection, Manual_edits	How cells were selected; manual boundary edits
QC_flags, QC_pass, Nuclei_in_cell, Nucleus_area, Nucleus_DAPI_mean	Quality-control flags and nuclear features
Nuc_Area, Nuc_Mean, Nuc_CTCF, Cyto_Area, Cyto_Mean, Cyto_CTCF	Nuclear and cytoplasmic measurements (6.5.1)
NC_ratio, Nuc_fraction, NC_flag, NC_margin_px	N/C ratio, nuclear fraction of signal, thin-cytoplasm flag, edge ring (pixels)
Fiji_check	Result of the Fiji verification, if done
App_version, Timestamp	Program version, date

9.3 Other outputs

- **CSV** (opens correctly in Excel, including non-ASCII characters).
- **ROI image (PNG):** with outlines and boxes.
- **ImageJ/Fiji comparison package (ZIP):** measured channel (TIFF) + ROIs (RoiSet.zip) + instructions (`HOW_TO_USE.txt`).

9.4 Not losing your work

- **Autosave:** after every addition, `Documents/CellVerity/otomatik_kayit.xlsx`. Even after a crash or power cut nothing is lost; on restart the program asks *"Continue where you left off?"*
- **Open previous results (Excel)...:** load a saved Excel file and continue measuring (the same image is not counted twice).
- Quitting with unsaved results asks whether to save.

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10. Plots and statistics (comparing groups)

Results ▸ Plot and statistics... (⌘G)

Setting	Options
Control group	Reference for Dunnett's test and normalisation (a group named *Control/Kontrol/untreated/vehicle*... is chosen automatically)
Variable	CTCF · CTCF_per_Area · Mean · IntDen · Area
Statistical unit	Cell · Image mean · **Experiment mean (recommended)**
Multiple comparisons	**Dunnett** (each group vs control) · Tukey HSD (all pairs)
Plot type	Bar (mean ± SEM) · Box · Violin — all with individual points
Group order	Drag to reorder (control always first, grey)
Normalise	Control mean = 1
Asterisks	GraphPad notation: ns, * p<0.05, ** p<0.01, *** p<0.001, **** p<0.0001

Tests: 2 groups → Welch's t-test + Mann–Whitney U. 3+ groups → one-way ANOVA + Kruskal–Wallis + Dunnett (or Tukey).

10.1 Important: the statistical unit (pseudoreplication)

Cells from the same experiment are not independent. Counting every cell as a replicate makes **p values look smaller than they are** (one of the most common issues flagged by reviewers). Example (simulated data, 3 experiments × 3 images × 6 cells):

Unit	n	Dunnett p (treatment vs control)
Cell	54	0.069
Experiment mean	3	**0.62**

Recommendation: in publications, run statistics on **means of independent experiments** (at least 3); individual cells can still be shown as points in the plot. The program warns when you test at the cell level.

Dunnett p values are computed by numerical integration; the program uses a fixed seed, so **the same data always give the same p**.

Save: plot as PNG (600 dpi), PDF, SVG; statistics as .txt.

10.2 Protein panel (several proteins, pathway level)

To see a drug's effect on a pathway, the proteins do **not** need to be measured in the same cell. In the same experiment, protein X can be measured in one set of wells and protein Y in others, each with a single channel.

1. When adding each image, type the protein in the **Protein / target** field of the right panel (e.g. *p53*). In batch mode, a `Protein/Experiment_1/Control/…` folder layout gives the protein from the folder name; otherwise one name for the whole folder can be entered in the dialog.
2. **Results ▸ Protein panel... (⌘⇧G).**
3. Different proteins are measured with different antibodies, so **raw CTCF values cannot be compared between proteins**. Each protein is normalised to **its own control in the same experiment** (fold change; from experiment means).
4. Plot: large dot = geometric-mean fold change, line = 95 % confidence interval, small dots = independent experiments, dashed line = control (1).
5. Test: one-sample t-test of $\log_2$(fold change) = 0 (equivalent to GraphPad's *ratio paired t test*), **Holm** correction across groups within each protein. n = independent experiments (at least 3 recommended).
6. An experiment without a control group is skipped and reported. Excel gets **Protein_panel** (summary) and **Protein_panel_experiments** (fold change per experiment) sheets.
7. With several proteins, the **Plots and statistics** dialog asks for a protein; group comparisons are done separately for each protein.

> ℹ️ This view shows how the proteins change together **at the pathway level** (e.g. does MDM2 rise when p53 rises with the drug?). It does not show the relationship within the same cell; that needs double staining.

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11. Fiji bridge and method validation

Fiji menu:

11.1 Verify with Fiji (⌘⇧F)

Has **Fiji's own Measure engine** measure the selected cells, the background boxes and (with DAPI) the **nuclear/cytoplasmic compartments in the background**. The compartments are exported as label images, Fiji creates its own selections (*Threshold → Create Selection*) and the N/C ratio is recalculated from Fiji's values. No Fiji window opens (~5 seconds); the results are shown side by side with the program's. When the image is added to the results table, the outcome is written to the **Fiji_check** column (e.g. *OK (1.54p99, max diff 3.3e-07 %)*). Changing the selection, boxes or boundaries invalidates the earlier check.

11.2 Open in Fiji

Opens Fiji with its interface: image, scale, all ROIs in the ROI Manager (cells, background boxes and, with DAPI, the `Nuc_n` / `Cyto_n` compartments), Measure results (ROI name on every row). Files: `Documents/CellVerity/Fiji/<image name>/`.

11.3 Compare with manually traced ROIs (method validation)

To quantify how close the automatic boundaries are to manual tracing (recommended for papers):

1. In Fiji, open the FITC image. Trace a cell with the **Polygon** or **Freehand** tool → **T** (add to ROI Manager). Repeat for all cells.
2. ROI Manager ▸ **More »** ▸ **Save...** → `manual_RoiSet_<image>.zip`
3. In the program, open the same image (and its DAPI) → **Fiji ▸ Compare with ROIs traced manually...** → choose the zip.
4. Each manual ROI is matched to the automatic cell it overlaps most: area difference, CTCF difference, **Dice** and IoU (boundary overlap; 1 = perfect).
5. ➕ **Add to validation table.** Repeat for 3–5 images (separate + touching cells, different conditions).
6. **Fiji ▸ Method validation table and Bland–Altman...:** scatter plot + Bland–Altman (percentage difference), mean difference (bias), 95% limits of agreement, Pearson r, Lin's CCC, Dice.

Recommendation: ≥ 30 cells, ≥ 3 images. Optionally re-trace the same cells a few hours later (intra-observer variability): if the automatic–manual difference is not larger than the manual–manual difference, the method is at least as reliable as manual measurement.

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12. Warnings: what they mean and what to do

12.1 Opening images

Warning	Meaning	What to do
The image appears to have been processed with software (Adobe Photoshop…)	The file was re-saved by an image editor	Use the raw microscope output
No camera information (EXIF)	Merged or re-saved image	Use the raw single-channel FITC
… looks like a DAPI image, open it as DAPI?	File colour does not match the button	Usually **Yes**
The FITC and DAPI images differ in size; DAPI was not used	The two files are not the same acquisition	Open the correct DAPI
New FITC opened; open the DAPI of the same field	The previous field's DAPI was removed	Open the new field's DAPI

12.2 Calibration and cell separation

Warning	Meaning	What to do
🔴 *The scale bar value could not be read; 20 µm was assumed*	Label not read	Type the value from the image into the **Scale bar** box
No scale bar found — pixel units	No scale bar	Area in pixel²; comparisons valid only at identical magnification
🔴 *Cell separation: shape only — N touching cell pairs*	No DAPI, touching cells present	Open the DAPI of the same field
🔴 *This DAPI image does not seem to belong to the same field*	Most nuclei lie outside the cells	Open the correct DAPI
Re-analysed: previous manual edits were reset	Scale etc. changed	Redo the edits

12.3 Background

Warning	Meaning	What to do
ℹ️ *All pixels in the boxes are 0*	Camera black level clipped; CTCF = IntDen	Fine if all images used the same settings

Warning	Meaning	What to do
⚠️ *Some boxes may contain signal/debris*	Bright pixels in a box	Drag the box to a clean area
🔴 *No background box — CTCF cannot be computed*	No room for boxes / no box in Manual mode	Reduce the box size or add boxes manually
⚠️ *The box overlaps a cell*	Boxes cannot be placed on cells	Place it a little further away

12.4 Measurement table

Situation	Meaning	What to do
Saturated px > 0	Pixels at the camera's maximum → CTCF is **underestimated**	Re-acquire with lower exposure, or exclude that cell

12.5 Statistics

Warning	Meaning	What to do
⚠️ *Each cell was counted as a replicate (risk of pseudoreplication)*	n = number of cells	Use the **Experiment mean** unit for publication
⚠️ *Different objectives are mixed (20×, 40×)*	Groups acquired at different magnifications	Compare only images from the same objective
⚠️ *The images were acquired with different exposure times*	Exposure differs (e.g. 0.5 s and 1 s) → CTCF values are not comparable	Compare only images with identical exposure/gain/lamp settings; untick the others in batch mode
⚠️ *Zero within-group variability*	Identical values repeat (same image added twice?)	Check the table; the test is not reliable
ℹ️ *2 experiments per group: low statistical power*	n < 3	At least 3 independent experiments
Statistics require at least … per group	Not enough data	Add more images/experiments
⚠️ *Few cells: r and CCC are meaningless* (validation)	Too few cells for Bland–Altman	≥ 30 cells, ≥ 3 images

12.6 Batch measurement and Fiji

Warning	Meaning	What to do
🟠 *no DAPI (best candidate …%)*	No overlapping DAPI found	Put the correct DAPI in the folder, or accept shape-only separation
⚪ *skipped: …*	See 8.2	—
Fiji was not found	Fiji not installed / elsewhere	Fiji ▸ Choose Fiji location…
⚠️ *Difference found* (Fiji verification)	Program and Fiji differ (not expected)	Check calibration; please report the issue

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13. For your paper: methods text and citation

Example methods text (adapt to your settings):

> Corrected total cell fluorescence (CTCF) was quantified as CTCF = integrated density – (cell area × mean background fluorescence) (McCloy et al., 2014) using CellVerity v2.2.0 (formerly ICC CTCF Ölçer; Balcan, 2026; https://doi.org/10.5281/zenodo.23884562), an open-source tool whose measurements are identical to ImageJ/Fiji for the same regions of interest (verified with Fiji 1.54p). Images were acquired on an Olympus BX43 fluorescence microscope with an Olympus DP74 camera (cellSens Entry); FITC fluorescence was measured on the green channel of the unprocessed images (40× objective, 0.145 µm/pixel; calibration read from the embedded scale bar). Cells were segmented from the FITC channel (Li threshold) and touching cells were separated by watershed seeded with DAPI-stained nuclei. Background was the mean of three 50 × 50 pixel cell-free regions per field. Cells touching the image border were excluded. Automatic boundaries were compared with manual Fiji tracing in n = … cells from … images (Bland–Altman bias …%, 95% limits of agreement … to …%; mean Dice …). Statistics were performed on the means of N = … independent experiments using one-way ANOVA followed by Dunnett's test versus control.

The **Methods** sheet in Excel and the *App_version, Nucleus_method, Scale_source, BG_box_px, BG_n, Selection, Manual_edits, Fiji_check* columns in the **Cells** sheet contain everything needed to fill in this text.

Citation (reference list):

> Balcan E. (2026). *CellVerity: verifiable per-cell fluorescence (CTCF) quantification for immunocytochemistry images* (Version 2.2.0) [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.23978391

DOI	When to use
DOI of the version you used (v2.2.0: 10.5281/zenodo.23978391)	Recommended for reproducibility — readers can download exactly the version you used
10.5281/zenodo.23978391	v2.2.0 (*CellVerity*)
10.5281/zenodo.23884562	v2.1.0 (*CellVerity*)
10.5281/zenodo.23860021	v2.0.0 (released as *ICC CTCF Ölçer*)
10.5281/zenodo.23860020	**All versions** of the program (concept DOI; always resolves to the newest version)

The version used is recorded in the **App_version** column in Excel; if you measured with a newer version, take that version's DOI from the Zenodo page (https://doi.org/10.5281/zenodo.23860020 → "Versions").

For the CTCF method: McCloy RA, Rogers S, Caldon CE, Lorca T, Castro A, Burgess A. (2014). Partial inhibition of Cdk1 in G2 phase overrides the SAC and decouples mitotic events. *Cell Cycle* 13(9):1400–1412. https://doi.org/10.4161/cc.28401

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14. Troubleshooting

Problem	Solution
Mac: *"unidentified developer"*	Right-click the app ▸ Open ▸ Open (once)
Mac: *"is damaged and can't be opened"*	Terminal: `xattr -cr "/Applications/CellVerity.app"`
Mac (Apple Silicon, Monterey): does not open	Update macOS to 12.3 or later
Windows: *"Windows protected your PC"*	More info ▸ Run anyway
Slow first start	A font cache is built on first launch (once, ~10–30 s)
Wrong cell count	Did you use the right button (FITC/DAPI)? Is the DAPI from the same field? Correct boundaries as in Section 7
A dense cluster counted as one "cell"	**Split** tool
Nuclei (magenta outlines) look wrong	Switch to **Display: DAPI** and check; is the DAPI from the same field?
Values differ slightly from v1	v2 uses different nucleus detection (Nucleus_method). Use one version throughout a study
Excel cannot be saved	If the file is open in Excel, close it and retry
The program closed / crashed	Reopen → *"Continue where you left off?"* → Yes (autosave)
Fiji bridge does not work	Fiji ▸ Choose Fiji location… ; open and close Fiji once yourself (first-run updates)
Interface in the wrong language	View ▸ Language / Dil, then restart

Questions and bug reports: https://github.com/erdal-b-dot/cellverity/issues

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15. Technical details

Measurement definitions (same as ImageJ): - Area = number of pixels × (µm/pixel)² - IntDen = Area × Mean (calibrated units); RawIntDen = sum of pixel values - StdDev = sample standard deviation (n – 1) - ROI rule: a pixel belongs to the ROI if its centre lies inside the polygon - CTCF = IntDen – Area × BG_Mean; CTCF_per_Area = Mean – BG_Mean

Calibration: the length of the cellSens scale bar (white box, lower right) is measured in pixels; the bar is located first (a thin black line with white above and below), the box is measured from the bar's white margin, and the number below it is read with digit templates. Tested on 3,534 fluorescence images (author's archive 3,233 + development set 301; two cellSens box styles): all read correctly. Objective = camera pixel size (DP74: 5.86 µm) ÷ µm/pixel.

Cell segmentation: FITC (G channel) Gaussian smoothing → Li threshold → closing, hole filling, small-object removal (< 50 µm²) → watershed seeded with DAPI nuclei (distance transform if no DAPI).

Nucleus detection (Nucleus_method): | Method | Version | Description | |---|---|---| | `otsu` | v1.0.0 | Otsu threshold (also includes the dim DAPI halo) | | `bright_core` | early v2 | Upper threshold of 3-class Otsu (misses dim nuclei) | | **`local`** | **v2.0.0 default** | Otsu mask → watershed from smoothed DAPI maxima → each nucleus trimmed at half of its own peak intensity |

Validation results: - Identical to Fiji 1.54p for the same ROIs: Area/Mean/StdDev/Min/Max/IntDen/RawIntDen (5 ROIs in a 40× image; 54 ROIs in a 20× image with non-zero background; largest difference < 0.00001 %). - Manual tracing (1 image, 2 touching cells — preliminary): Dice 0.943 / 0.938; CTCF difference +2.3 % / +3.7 %. - Dense clusters at 20×: 47 nuclei, 51 cells detected.

Program files: `core.py` (measurement), `batch.py` (batch), `report.py` (Excel, plots, statistics), `fiji.py` (Fiji bridge), `desktop.py` (interface), `i18n.py` + `locales/` (languages), `tests/test_core.py` (validation tests).